# Welcome To

# SECP2523
# DATABASE

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# Database Application Life Cycle

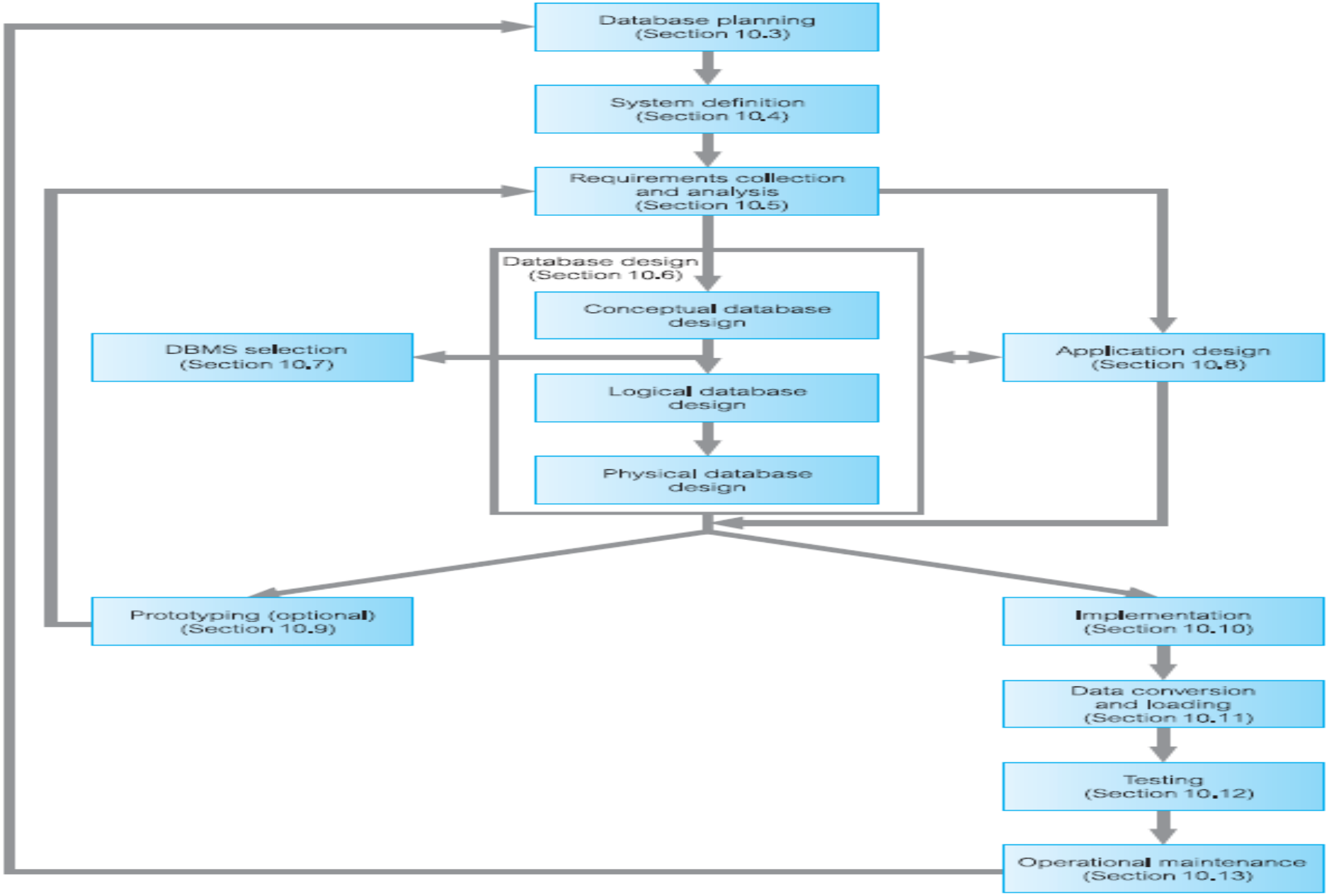
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# Information System Development Lifecycle

- A computer-based information system includes a database, database software, application software, computer hardware, and personnel using and developing the system.
- The database is a fundamental component of an information system, and its development and usage should be viewed from the perspective of the wider requirements of the organization.
- Therefore, the lifecycle of an organization's information system is inherently linked to the lifecycle of the database system that supports it.
- The stages in the lifecycle of an information system include: planning, requirements collection and analysis, design, prototyping, implementation, testing, conversion, and operational maintenance.

# Stages of Database Application Lifecycle

As a database system is a fundamental component of the larger organization-wide information system, the database system development lifecycle is inherently associated with the lifecycle of the information system



# Database Planning

- Management activities that allow stages of database application lifecycle to be realized as efficiently and effectively as possible.
- Must be integrated with overall IS strategy of the organization.
- Include development of standards that govern:
  - how data will be collected,
  - how the format should be specified,
  - what necessary documentation will be needed,
  - how design and implementation should proceed.

# Database Planning: Mission Statement

- defines major aims of database application.
- clarify purpose of the database project and provides clearer path towards the efficient and effective creation of required database system

"The purpose of the *DreamHome* database system is to maintain the data that is used and generated to support the property rentals business for our clients and property owners and to facilitate the cooperation and sharing of information between branches."

# Database Planning: Mission Objectives

- Each objective should identify a particular task that the database must support

To maintain (enter, update, and delete) data on branches.
To maintain (enter, update, and delete) data on staff.
To maintain (enter, update, and delete) data on properties for rent.
To maintain (enter, update, and delete) data on property owners.
To maintain (enter, update, and delete) data on clients.
To maintain (enter, update, and delete) data on property viewings.
To maintain (enter, update, and delete) data on leases.
To maintain (enter, update, and delete) data on newspaper adverts.

To perform searches on branches.
To perform searches on staff.
To perform searches on properties for rent.
To perform searches on property owners.
To perform searches on clients.
To perform searches on property viewings.
To perform searches on leases.
To perform searches on newspaper adverts.

To track the status of property for rent.
To track the status of clients wishing to rent.
To track the status of leases.

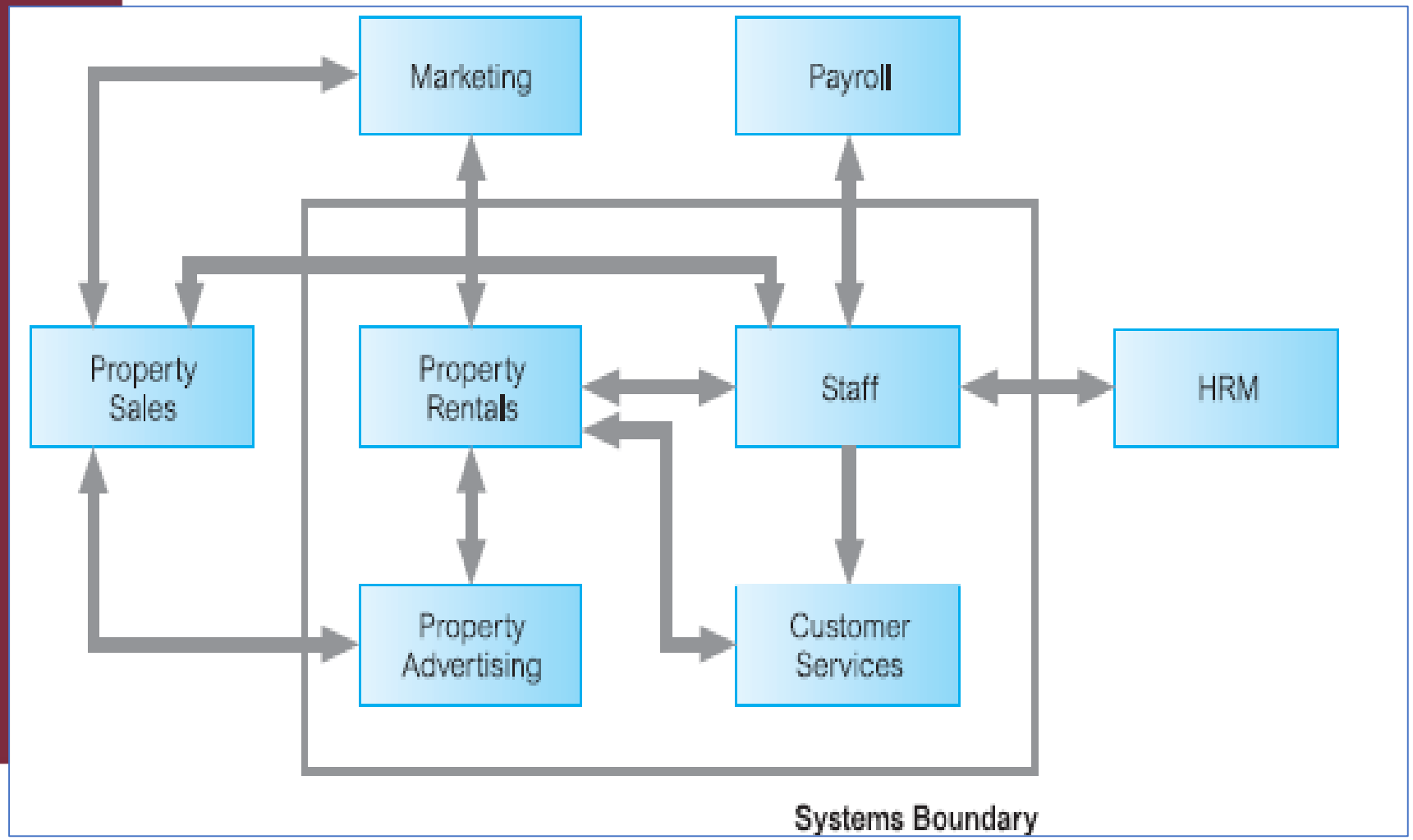
To report on branches.
To report on staff.
To report on properties for rent.
To report on property owners.
To report on clients.
To report on property viewings.
To report on leases.
To report on newspaper adverts.

**Figure 11.9** Mission objectives for the *DreamHome* database system.

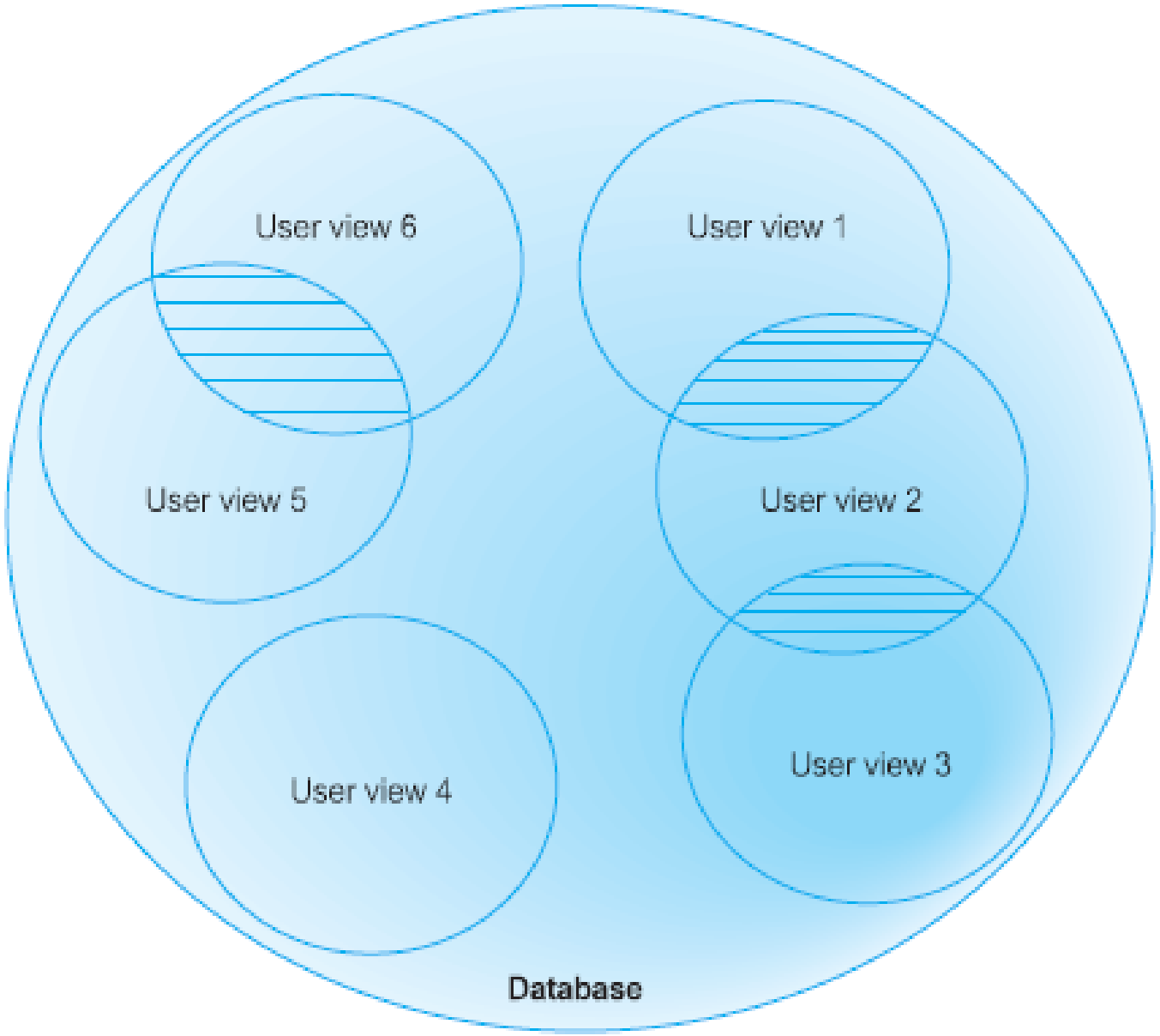
# System Definition

- Describes **scope** and **boundaries** of database application and the major user views.
  - System boundaries should involve current and future application areas and users
- **User view** defines what is required (user requirement) of a database application from the perspective of:
  - a particular job role (such as Manager or Supervisor) – analyst must identify user views
  - enterprise application area (such as marketing, personnel, or stock control)

# System Definition: System Boundary



# System Definition: Multiple User Views




Database system

# Requirements Collection and Analysis

- Collecting and analysing information about the part of organization to be supported by the database application, and using this information to identify users' requirements of new system.

- Information is gathered for each major user view including:
  - A description of data used or generated;
  - Details of how data is to be used/generated;
  - Any additional requirements for new database application.
  - Documented in a Requirements Specification document

# Requirements Collection and Analysis

- Features in requirement specification:
  - initial database size;
  - database rate of growth;
  - the types and average number of record searches;
  - networking and shared access requirements;
  - performance;
  - security;
  - backup and recovery;
  - legal issues.

***Initial database size***

(1) There are approximately 2000 members of staff working at over 100 branches. There is an average of 20 and a maximum of 40 members of staff at each branch.

(2) There are approximately 100,000 properties available at all branches. There is an average of 1000 and a maximum of 3000 properties at each branch.

# Requirements Collection and Analysis: Requirement Specification

### *Database rate of growth*

(1) Approximately 500 new properties and 200 new property owners will be added to the database each month.

### *The types and average number of record searches*

(1) Searching for the details of a branch—approximately 10 per day.

(2) Searching for the details of a member of staff at a branch—approximately 20 per day.

### *Networking and shared access requirements*

All branches should be securely networked to a centralized database located at *DreamHome's* main office in Glasgow. The system should allow for at least two to three people concurrently accessing the system from each branch. Consideration needs to be given to the licensing requirements for this number of concurrent accesses.

### *Performance*

(1) During opening hours, but not during peak periods, expect less than a 1-second response for all single record searches. During peak periods, expect less than a 5-second response for each search.

# Requirements Collection and Analysis: Requirement Specification

***Security***

(1) The database should be password-protected.

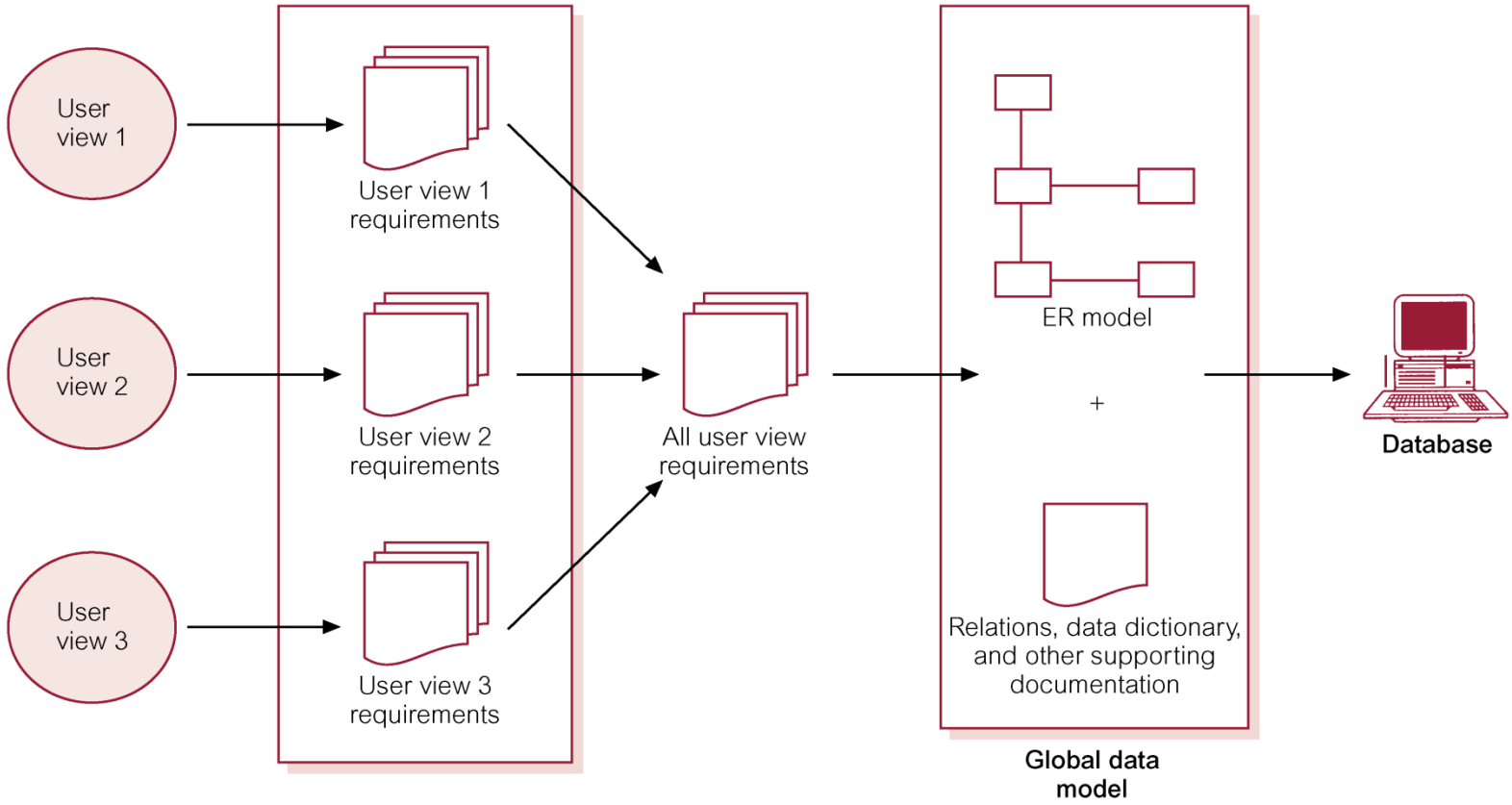
(2) Each member of staff should be assigned database access privileges appropriate to a particular user view, namely Director, Manager, Supervisor, or Assistant.

**Backup and Recovery**

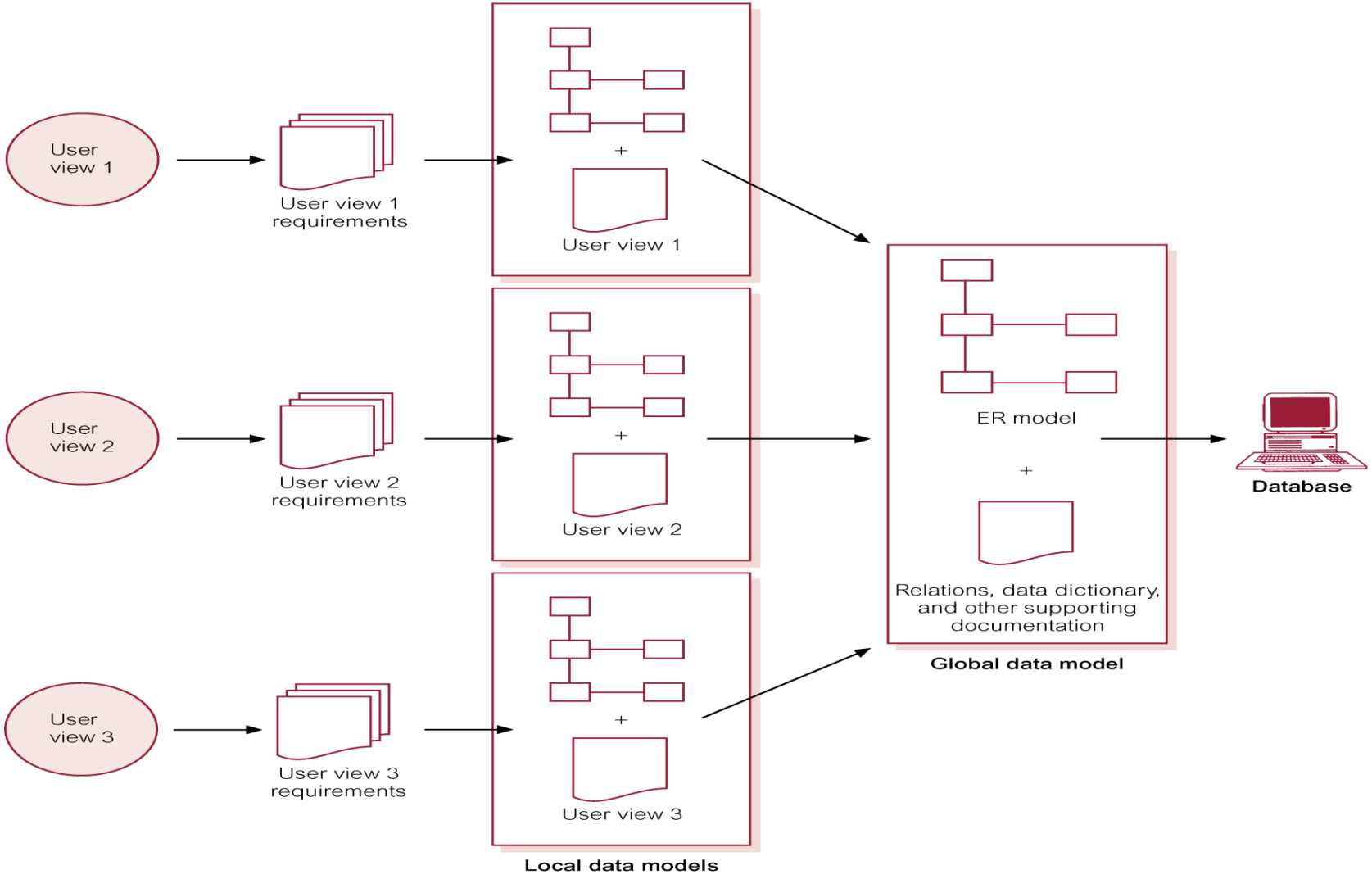
The database should be backed up daily at 12 midnight.

# Requirements Collection and Analysis: Centralized Approach

- 3 main approaches:
  - Centralized
  - View Integration
  - Combination of both approaches



# Requirements Collection and Analysis: View Integration



# Database Design: Definition

- The process of creating a design for a database that will support the enterprise's operations and objectives.

- Major aims:
  - Represent data and relationships between data required by all major application areas and user groups.
  - Provide **data model** that supports any transactions required on the data.
  - Specify a minimal design that is appropriately structured to achieve stated performance requirements for the system (such as response times).

# Database Design: Approaches

- **Top-down**
  - Starts with developing data models which contain high-level entities and relationships
  - Then refine this high-level data models into a lower-level entities, relationships and attributes
  - Use **Entity-Relationship model**
- **Bottom-up**
  - Begins at the fundamental analysis of the associations between attributes
  - Grouped into relations that represent types of entities and relationship between entities
  - Use **Normalization process**
- **Inside-out**
  - Begins with identifying major entities
  - Then spreading out to consider other entities, attributes associated to those first identified
- **Mixed**

# Database Design: Why data modeling

- Main purposes of data modeling include:
  - to assist in understanding the meaning (semantics) of the data;
  - to facilitate communication about the information requirements.
- Building data model requires answering questions about entities, relationships, and attributes.
  - A data model ensures we understand:
    - each user's perspective of the data;
    - nature of the data itself, independent of its physical representations;
    - use of data across user views.

# Database Design: DB Design Phase

**Conceptual**

- Process of constructing a **model of information** used in an enterprise, independent of all physical considerations.
- **Entity Relationship Model; Data dictionary**

**Logical**

- Process of constructing a model of information used in an enterprise based on a specific data model (e.g. relational), but independent of a particular DBMS and other physical considerations.
- **Normalized database (relational) schemas**

**Physical**

- Process of producing a description of the implementation of the database on secondary storage; it describes the base relations, file organizations, and indexes design used to achieve efficient access to the data, and any associated integrity constraints and security measures.

# Database Lifecycle: Other Stages

5. DBMS selection
6. Application design – user interface design and application program design
7. Prototyping
8. Implementation
9. Data Conversion and loading
10. Testing
11. Operational Maintenance

# Summary of DB Lifecycle

| STAGE | MAIN ACTIVITIES |
|---|---|
| *Database planning* | Planning how the stages of the lifecycle can be realized most efficiently and effectively. |
| *System definition* | Specifying the scope and boundaries of the database system, including the major user views, its users, and application areas. |
| *Requirements collection and analysis* | Collection and analysis of the requirements for the new database system. |
| *Database design* | Conceptual, logical, and physical design of the database. |
| *DBMS selection* | Selecting a suitable DBMS for the database system. |
| *Application design* | Designing the user interface and the application programs that use and process the database. |
| *Prototyping (optional)* | Building a working model of the database system, which allows the designers or users to visualize and evaluate how the final system will look and function. |
| *Implementation* | Creating the physical database definitions and the application programs. |
| *Data conversion and loading* | Loading data from the old system to the new system and, where possible, converting any existing applications to run on the new database. |
| *Testing* | Database system is tested for errors and validated against the requirements specified by the users. |
| *Operational maintenance* | Database system is fully implemented. The system is continuously monitored and maintained. When necessary, new requirements are incorporated into the database system through the preceding stages of the lifecycle. |

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